

SUSHANT LOKHANDE

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EXPERIENCE

San Jose State University | Graduate Research Assistant, Machine Learning Systems Aug 2025 to May 2026

- Automated provisioning, configuration, and **telemetry** for containerized test clusters, cutting experiment setup effort **55%** and making every sweep reproducible from config.
- Boosted malware classification to **80.2%** across 17 families with a soft-voting **PyTorch** ensemble (CNN, ResNet50, ViT), **2.4 pts** over the best single model at $p < 0.001$.
- Built an automated evaluation harness for controlled experiments, cutting setup effort **55%** and making every result reproducible from a single config.

Turito | Software Engineering Intern, Systems & Performance Jun 2022 to Oct 2022

- Slashed p95 **CPU 22%** by profiling high-frequency media paths and eliminating excess buffer allocation, payload copies, repeated routing lookups, and **lock contention**.
- Cut controlled-disruption failures **42%** (3.1% to 1.8%) by hardening **retries**, timeouts, failure recovery, and distributed state transitions.

Trivia Softwares | Software Engineering Intern Nov 2021 to Jan 2022

- Built a **Rust/Tokio** load-testing framework driving about **25,000** concurrent sessions, exposing lock contention, serialized **WebSocket** writes, and slow session cleanup.
- Raised equivalent-load signaling throughput about **31%** by surfacing concurrency bottlenecks that drove targeted backend fixes.

TECHNICAL SKILLS

Languages: Python, C++17, Rust, TypeScript, SQL

ML & Retrieval: PyTorch, transformers, embeddings, RAG with citations and abstention, two-tower retrieval, HNSW and ANN search, semantic caching

Evaluation: LLM eval harnesses, McNemar significance testing, leakage-free temporal splits, Recall/NDCG/MAP, failure classification, human-review gates

Systems: FastAPI, Kafka, Temporal, gRPC, Docker, Kubernetes, AWS, Linux, CI/CD

PROJECTS

Proxima, Vector Search Engine | C++17, AVX2/SIMD, HNSW, pybind11

- Reached **0.999 recall** at **12,049 QPS** single-threaded, **1.8x** hnswlib and **2.5x** FAISS, with an HNSW engine written from scratch in C++17.

AI Learning Hub, Grounded RAG | Python, FastAPI, LangChain, OpenAI, Docker, AWS

- Grounded answers over **10,000+** articles with source citations and abstention when retrieved evidence was insufficient.

ReelRank, Two-Stage Recommender | PyTorch, C++ HNSW, FastAPI, React, TypeScript

- Built a **two-tower** retrieval model with ANN candidate generation and a neural ranker, evaluated on a leakage-free temporal split with Recall, NDCG, and MAP.

OPEN SOURCE

- microsoft/markitdown**: fixed EPUB conversion for percent-encoded manifest paths, with regression tests (merged)
- NVIDIA/garak**: preserved attempt UUIDs in report aggregation for traceable LLM evals (merged, shipped in v0.17.0)

EDUCATION

San Jose State University | M.S. in Computer Science | GPA 3.8/4.0 | Spartan Scholar Aug 2024 to May 2026

Coursework: Machine Learning, Artificial Intelligence, Advanced Algorithms, Distributed Computing, Database Systems, GPU Programming

University of Mumbai | B.Tech. in Computer Engineering | GPA 3.9/4.0 Jun 2019 to May 2023

Research: malware image-classification ensemble selected for Dr. Mark Stamp's ML textbook; sole-author contactless palmprint verification (0.9975 AUC)